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3D printed high–strength polyimide aerogel metamaterials for sound absorption and thermal insulation

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ABSTRACT

The structural manufacturing of high–performance polyimide (PI) aerogels is challenging because of the insufficient mechanical strength and rheological properties of sol–gel ink, and PI aerogels have limited application. In this study, a PI aerogel–based Helmholtz resonator (PIHM) and its acoustic metamaterial were innovatively constructed using freeze casting–assisted extrusion printing process and building block–assembly strategy, featuring a micro–perforated panel (MPP) sound–absorbing structure. The PIHM with a density of 0.294 g·cm^{-3} achieved a compressive strength of 10.2 MPa because of its periodically distributed honeycomb topology. Moreover, the PIHM not only retained the lightweight and thermal insulation characteristics of aerogels but also exhibited excellent sound–absorption performance because of the dual dissipation of sound waves by the MPP structure and PI aerogel framework. By serially connecting PIHMs with different aerogel pore sizes and leveraging the distinct acoustic properties of the layered structure along with the significant increase in relative mass resistance and acoustic impedance, the acoustic metamaterial achieved absorption coefficient peaks of 0.82–0.91 at 622–726 Hz, considerably widening the bandwidth with an absorption coefficient greater than 0.8. Finally, the composite sound–absorbing panel fabricated from a PIHM combined with common building materials demonstrated strong practicability and versatility. This research has pioneered a viable method for manufacturing PI aerogels with functional structures through 3D printing, expanding their application in the field of sound absorption and thermal insulation, and paving the way for the study of aerogel metamaterials in construction.

1. Introduction

Acoustic metamaterials are artificially engineered materials with unique capabilities for manipulating sound waves, achieved through the intricate design of their microstructures, which enable unusual propagation characteristics [1]. In the field of architecture, the application of acoustic metamaterials is primarily focused on sound absorption and noise reduction, encompassing various structural types such as Helmholtz resonator architecture, membrane-based structures, space coiling acoustic metamaterials, gradient index acoustic metamaterials, and tunable acoustic metamaterials [2]. Among these, the micro–perforated panel (MPP) sound–absorbing structure based on the Helmholtz resonator possesses a resonant structure with low acoustic mass and high acoustic impedance, consisting of a thin plate with perforations and a connected back cavity [3]. Its sound–absorption bandwidth is primarily determined by the mass reactance and acoustic impedance. Absorption bandwidth increases with decreasing mass reactance. However, reducing the mass reactance by decreasing the thickness of the perforated plate or increasing the perforation rate reduces the sound resistance of the perforated plate, thereby affecting the resonance absorption peak. Consequently, some scholars have exploited the resistive sound–absorption characteristics [4], using porous materials in the back cavity or forming the back cavity out of porous materials to achieve broadband and low–frequency sound–absorption effects [5–7]. Among various porous materials, polyimide (PI) aerogels not only have small pores and high porosity but also possess excellent mechanical properties and wear resistance [8]. However, the performance–oriented structural manufacturing of aerogels through traditional processes is challenging. Common moulds cannot completely block shapes, such as cubes or cylinders, and demoulding can easily damage samples. To address these issues, aerogel manufacturing through 3D printing has emerged as a new research direction.

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Direct ink writing (DIW) [9–11], which is based on material extrusion, is one of the primary methodologies for 3D aerogel printing. This technique features a straightforward architecture employing a feed cylinder, extrusion mechanism, nozzle, printing platform and electronic control system. The printing operation involves air pressure modulation, piston displacement or screw rotation speed to extrude viscoelastic ink from the nozzle, enabling layering in a bottom–up fashion along a predetermined trajectory for the swift fabrication of 3D constructs. A crucial aspect of this process is the ink's requisite rheological property [12], which not only facilitates the smooth extrusion of high–viscosity ink but also enables a printed structure to support itself. However, system errors in DIW results in 'stringing' and 'distortion' after the ink is extruded from the nozzle, affecting the moulding effect of printed samples. For PI aerogels, the precursor polyamic acid sol has shear–-thinning properties but at low shear stress [13]. The loss modulus is usually higher than the storage modulus. The commonly used curing agents are acetic anhydride and pyridine, which are toxic substances that pose environmental risks, and the rapid sol–gel curing rate not only gradually increases ink viscosity and complicates extrusion processes but also limits the print size of samples. Therefore, we proposed a method that utilises a building block–assembly strategy and freeze casting to complement DIW for PI aerogel production without the addition of rheology modifiers [14]. This method successfully printed pure PI aerogels, overcoming challenges associated with printing errors and enabling the manufacture of large components.

To fully leverage the functional structure design advantages of 3D printing, we integrated the MPP sound–absorbing structure with aerogel to fabricate a PI aerogel–based Helmholtz resonator (PIHM). A novel metamaterial was fabricated by serially connecting PIHMs with aerogels with pore sizes and porosities. The performance of the PIHM can be tuned by altering the honeycomb side length and ink viscosity. Within a density range of 0.153–0.294 $g\cdot cm^{-3}$, the thermal conductivity remained between 0.0317 and 0.0665 $W\cdot m^{-1}\cdot K^{-1}$, and the compressive strength increased from 5.45 MPa to 10.2 MPa, demonstrating exceptional mechanical properties and good thermal insulation characteristics. Additionally, the trend in sound–absorption performance aligns with the acoustic theory of MPP. Owing to the varying acoustic properties of different layered structures and the significant increase in relative mass resistance and acoustic impedance, the acoustic metamaterial of the PIHM successfully lowered the sound–absorption frequency and broadened the absorption bandwidth. Ultimately, we fabricated high–strength composite with sound–absorbing panels and PIHM combined with aluminium and lightweight wood. This PIHM demonstrated a simple and universal strategy for practical applications. The findings indicate that the lightweight, thermal–insulating, sound–absorbing and compressive PIHM can be used to provide thermoacoustic protection in many fields, such as construction, transportation and defence. The serially connected acoustic metamaterials exhibit broadband and low–frequency sound–absorption characteristics, holding significant research and application value in the field of acoustics.

2. Experimental section

2.1. Materials

4,4′–Diaminodiphenyl ether (ODA, 98 %), 4,4′–Diphenoxydianiline (ODPA, 97 %), N–Methyl–2–pyrrolidone (NMP, AR, >99 %), and triethylamine (TEA, AR, 99 %) were procured from Macklin Technology Co., Ltd., Shanghai, China. Deionized water was obtained using an ultrapure water instrument provided by Pingguan Technology Co., Ltd., Hubei, China. All the aforementioned reagents were used as received without any further treatment.

2.2. Preparation of printing ink

10.01 g ODA was dissolved in 113.734 ml NMP, followed by the addition of 15.655 g ODPA. The mixture was stirred for 6 h in an ice bath and then precipitated in deionized water to obtain PAA filaments. The washed PAA was subsequently cut into 1–3 mm lengths and freeze–dried for 72 h at 30 ℃. After drying, it was placed in 102 ml of deionized water and 10 ml of TEA added, followed by stirring for 6–8 h. Finally, the solution was degassed using a vacuum deaerator with a rotation speed of 1000–1500 rpm and a degassing time of 10–20 min to obtain PAAS sol ink suitable for printing. Different ink viscosity was obtained by controlling the amount of ODA and ODPA.

2.3. 3D printing of the PIHM

The experimental custom 3D printer consists of a DIW printing system and a refrigeration unit (Zhezheng Intelligent Technology Co., Ltd., Sichuan, China), and screw extrusion is used. Before printing, a print model is designed in SolidWorks (version 2022) and imported into Simplify 3D for slicing. Then, print parameter design is performed with the slicing software. The extrusion speed is set at 10–12 $mm\cdot s^{-1}$, the extrusion ratio is 1.0 and the nozzle diameter is 0.8 mm. The movement speed for the X, Y and Z axes is maintained at 35 $mm\cdot s^{-1}$. Subsequently, the refrigeration unit is activated to reduce the print platform temperature to −40–20 °C, and the surrounding ambient temperature does not exceed 20 °C. During printing, the distance between the nozzle and freezing platform is adjusted via the screw to remain at 0.8 mm, and Simplify 3D is used in monitoring printing progress. After printing, a building block–assembly strategy is employed to join the MPP, back cavity and base plate. The assembled sample is then placed in a freezer (Yaxingyike Technology Co., Ltd., Beijing, China) for secondary freezing at −80 °C. Once fully solidified, the sample is transferred to a freeze dryer (Yaxingyike Technology Co., Ltd., Beijing, China) for drying at temperatures of −50–30 °C over a period of 48–72 h. After drying, the sample is placed in a muffle furnace (Tianjingzhonghuan Electric Furnace Co., Ltd., Tianjing, China) and imidised at 200–250 °C for 3–5 h. This process culminates in the production of the 3D–printed PIHM.

2.4. Preparation of PIHM acoustic metamaterial

The assembly method employed is the building block–assembly strategy. Initially, two samples that have undergone secondary freezing and solidification are end–to–end joined on a freezing platform (one of which lacks a base plate). To ensure optimal joining, the freezing platform temperature is set at −40 °C, and the surrounding ambient temperature maintained below 20 °C. Subsequently, a heat gun (GHG20–63, Bosch, Germany) is used to apply heat to the joint. The hot–air temperature is 40 °C, the airspeed is 5 $m\cdot s^{-1}$, the distance is 3–5 cm and the heating duration is less than 1 min. After resting for 10 mins, the assembled sample is placed in a −80 °C freezer for freezing and solidification. Finally, the PIHM acoustic metamaterial is obtained through freeze drying and thermal imidisation.

2.5. Characterisation instruments and methods

The volumetric density of the sample is measured using Archimedes's principle of water displacement. The procedure is as follows: first, the dry sample is weighed, and the mass is recorded as M_0; then, the sample is submerged in a beaker filled with deionised water and soaked under vacuum conditions for 5–10 mins. The sample is then removed and weighed. The weight is denoted as M_1; subsequently, the surface water of the sample is wiped off and the sample is weighed again. This weight is recorded as M_2. The density of the deionised water is denoted as ρ_1. Finally, the volumetric density ρ is calculated using the formula:

$$\rho = \frac{M_0 \rho_1}{M_1 - M_2}, \# \tag{1}$$

The shrinkage rate of the sample refers to the ratio of the dimensions of the sample after imidisation to the dimensions after freeze solidification. The diameters and height of the sample in both states are measured using a vernier caliper. The axial and radial shrinkage rates are then calculated using the following formula:

$$\xi = \frac{l_a - l_b}{l_a} \times 100\%, \# \tag{2}$$

where l_a represents the sample dimensions after freeze solidification and l_b denotes the dimensions after imidisation. The microstructure of the sample is characterised using a scanning electron microscope (GeminiSEM 560, ZEISS, Germany). Before imaging, the samples are sputter coated with gold in an ion sputtering coater (SC7620, Phenom World, China) to ensure clear morphology under the electron microscope. The testing acceleration voltage is 15 KV. The pore size distribution of the sample is measured using a high–performance automatic mercury intrusion porosimeter (AutoPore 9600, Micromeritics, America), and the distribution of different pore sizes in the material is determined with the mercury intrusion method. Testing parameters include a sample size not exceeding 15 × 15 × 15 mm^3, sample weight of 2–3 g and pressure range from 3.4 × 10^3 Pa to 2.3 × 10^8 Pa. The infrared spectrum of the sample is analysed using a Fourier transform infrared spectrometer (Nicolet iS20, Thermo Scientific, America). The wavenumber range is 400–4000 cm^{-1}, 32 scan repetitions are performed and the resolution is 4 cm^{-1}. The thermal conductivity of the sample is measured using a transient hot–wire thermal conductivity metre (TC3200, XIATECH, China). The testing method is the transient hot–wire technique, adhering to the standard GB/T 34336–2017. Specifically, two samples with diameters of 30 mm and height of 10 mm are used to press against the testing probe, and three sets of data are recorded at room temperature (25 °C). The average value taken as the final measurement result. TGA is conducted using a thermogravimetric analyser (HP–TGA 750, TA Instruments, America). The TGA curve of each sample in an air atmosphere is obtained over a temperature range from room temperature to 800 °C and heating rate of 10 °C/min. The infrared thermal image and temperature curves are obtained by the infrared thermal imager (FOTRIC 285 s, FOTRIC INC, China). The temperature measurement accuracy is ±2 °C, and the heating platform temperature is 150 °C. The mechanical properties of PIHM are characterised using an electronic universal testing machine (CMT6103, MTS, America). The test samples are cylindrical, with diameters of 20 mm and height of 20 mm. The machine compresses at a speed of 0.5 mm·s^{-1} and stops at a compressive strain of 70 %. The force is unloaded, and the stress–strain curve is obtained. The sound–absorption performance of the sample is measured using a transfer function sound insulation measurement system (BK 4206, HBK, Britain). The testing method is the transfer function technique for characterising sound–absorption performance according to the measured transfer function of sound waves before and after sound passes through the aerogel sample. The sample diameter for the test is 29 mm, the frequency range is 200–6.4 kHz and the test is conducted according to the standard GB/T 18696.2–2002.

3. Results and discussion

3.1. Preparation of PIHM and its acoustic metamaterial

PIHM was fabricated through freeze casting–assisted DIW printing. The ink formulation and aerogel printing are identical to those used in the preparation of HPI aerogels [14]. By substituting the temperature field for rheology modifiers, no additives are not needed in the sol–gel ink. Thus, the rheological requirements of the ink are reduced, and any impact on the properties of the PI aerogel is prevented. In the printed HPI aerogels, the storage modulus (G′) of the polyamic acid salt (PAAS) sol–gel ink is always lower than the loss modulus (G″), exhibiting no yield point and maintaining shear–thinning properties. That is, the ink is easily extruded during printing. However, the hexagonal honeycomb structure limits the viscosity of the sol–gel ink because an increased number of intricate movements by the print head can lead to 'stringing' when extremely viscous ink is reduced. 'Stringing' results in poor moulding effects or failure. Conversely, extruded ink with extremely low viscosity is prone to collapse or point–like solidification after leaving the nozzle. Therefore, regulating ink viscosity is essential to successful moulding. Ink viscosity suitable for printing hexagonal honeycomb back cavities range from 10 Pa·s to 200 Pa·s. Within this range, the performance characteristics of the PI aerogels are linearly related to viscosity. In order to intuitively reflect the changes in the morphology and properties of the material, ink viscosity of 20, 96 and 182 Pa·s are selected in this study.

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Fig. 1a–b illustrates the structural schematic and printing process of the PIHM (Video S1, Supporting information). The PIHM consists of three layers, each of which is fabricated by PI aerogel printing. The top layer is an MPP, and the middle part comprises hexagonal honeycomb back cavities. Fig. 1c shows back cavity samples with side length of 2, 3 and 4 mm, respectively. The bottom layer is a solid base plate. The entire framework exhibits the structural characteristics of PI aerogel and is characterised by a surface distribution of micropores with adjustable porosity and pore size. Owing to the precision limitations of the printing method, the minimum controllable size of the perforation diameter is 1 mm, meeting sub–millimetre–level requirements. Additionally, to minimise the impact of the back cavity framework wall thickness on sound–absorption performance, the extrusion multiplier in the printing parameters is kept constant. After printing, the MPP, back cavity and base plate are assembled and joined with a building block–assembly strategy followed by secondary freezing, supercritical drying and thermal imidisation for the preparation of the PIHM. Fig. 1d shows the schematic structure of the PIHM acoustic metamaterial. A building block–assembly strategy is used to serially connect two PIHM blocks. By controlling the ink viscosity, the different acoustic characteristics is obtained by regulating the aerogel structure of the upper and lower layers. This method achieves broadband and low–frequency sound–absorption effects.

3.2. Structural characteristics of PIHM

The PIHM is printed using a 0.8 mm nozzle. The MPP has a pore diameter of 1 mm, the back cavity wall thickness is 0.8 mm and the solid base plate thickness is 1 mm (Fig. 2a). In practical applications, the external structure and material properties

can be controlled by adjusting printing parameters [15–17]. Here, PIHM–20, PIHM–96 and PIHM–182 correspond to samples prepared using ink with viscosities of 20, 96 and 182 Pa·s, respectively. Given that the ink viscosity is the same as that of the HPI aerogel, the microstructure of the aerogel remains consistent, that is, the aerogel pore

size and porosity decrease with increasing ink viscosity. When the ink viscosity increases from 20 Pa·s to 182 Pa·s, the aerogel pore size decreases from 11.2 μm to 5.2 μm (Fig. 2b–e) because as the solid content in the ink increases, the growth of ice crystals obtained through the freeze–casting method is inhibited. Concurrently, the porosity of the aerogel decreases from 82 % to 60 %. Therefore, the microstructure of the PIHM can be altered by controlling the ink viscosity. Fig. 2f depicts the chemical structural characteristics of PIHM at different ink viscosities. All three types of samples exhibit the same imide characteristic peaks. A peak at 1772 cm^{-1} represents the asymmetric stretching vibration of C═O, and the peak at 1723 cm^{-1} is attributed to the symmetric stretching vibration of C═O, the peaks at 1503 and 1375 cm^{-1} respectively represent the stretching vibrations of C═C and C–N–C and the peak at 748 cm^{-1} indicates the bending vibration of C═O. The

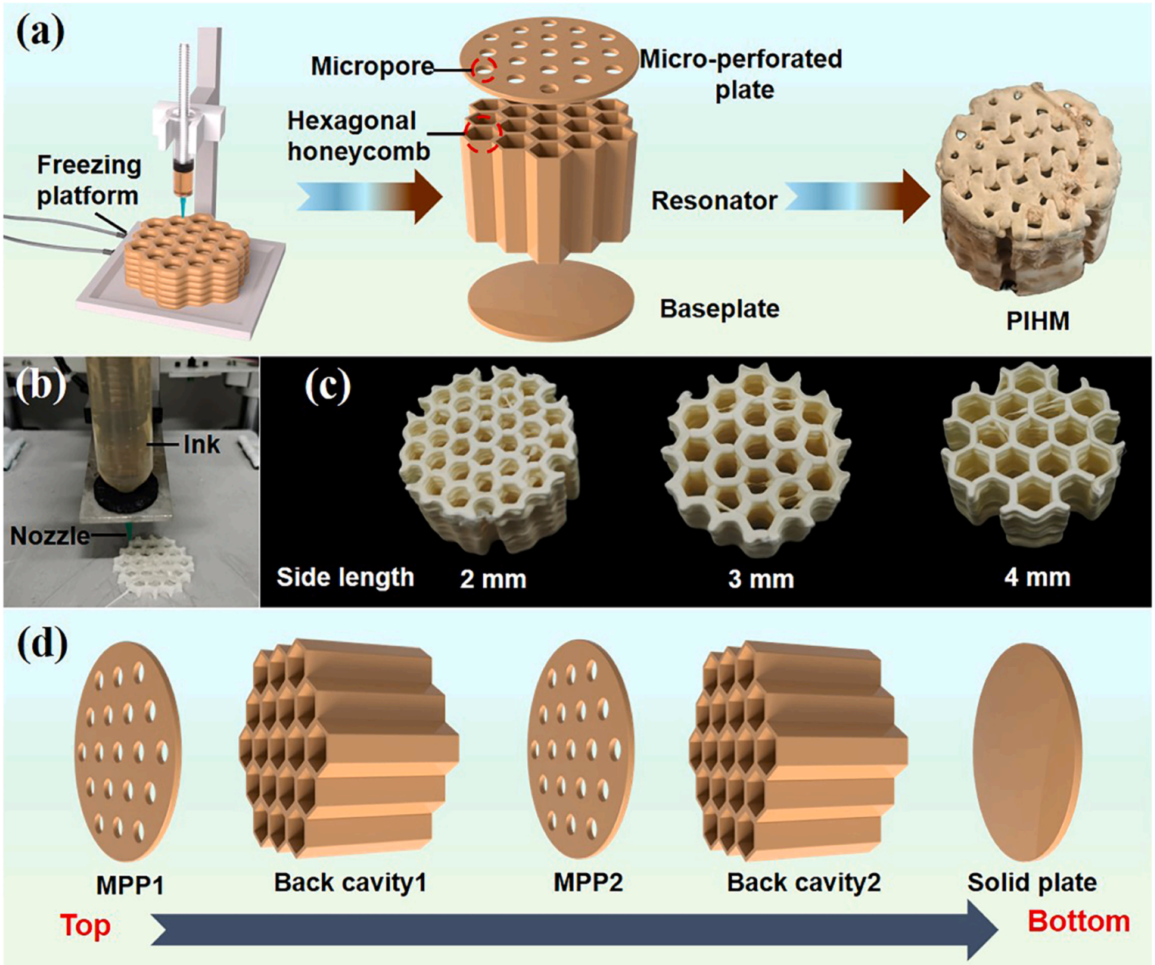

Fig. 1. Fabrication of PIHM and its acoustic metamaterial. (a–b) Schematic diagram and printing process of PIHM. (c) Back cavities with different honeycomb side lengths after freeze drying. (d) Assembly schematic of the PIHM acoustic metamaterial.

absence of amide II band or C=O stretching vibration peaks near 1540 and 1660 cm^{-1} indicate that PI imidisation is complete.

3.3. Density and shrinkage characteristics of the PIHM

A honeycomb structure with periodic topological distribution characteristics possesses a higher macro porosity and lower mass density than conventional aerogel materials. Table 1 lists the typical physical properties of the PIHM. The density of the PIHM increases with ink viscosity. As the ink viscosity increases from 20 Pa·s to 182 Pa·s and the honeycomb side length decreases from 4 mm to 2 mm, the density increases from 0.153 g·cm^{-3} to 0.294 g·cm^{-3}, indicating the lightweight nature of the structure (Fig. 2g). Moreover, the number of imide rings in the PAAS increases with ink viscosity. During thermal imidisation, the sample exhibits large dehydration shrinkage, which leads to increases in radial and axial shrinkage rates. Similarly, the number of radial honeycomb frameworks increases with decreasing honeycomb side length, and this effect results in considerable dehydration shrinkage and increase in the radial shrinkage rate. Conversely, the axial shrinkage rate is unrelated to the honeycomb side length.

The thermal conductivity is represented as the average value ± standard deviation from three independent experiments.

3.4. Mechanical properties of the PIHM

Each cell of the honeycomb structure is interconnected through nodes, creating an exceedingly stable three–dimensional network. When subjected to compression, each cell absorbs energy through the bending deformation of the cell walls and the plastic hinges at the wall junctions. Meanwhile, the nodes transfer the pressure to adjacent cells, effectively dispersing and redistributing the stress, which consequently enhances the structure's compressive strength [18]. Fig. 3a shows that the PIHM weighing 2.74 g can easily support a weight 1825 times its own. When the density is 0.153 g·cm^{-3}, the compressive strength is 5.45 MPa (obtained from the linear portion of the strain from 0 % to 10 %; Fig. 3b). The compressive modulus increases with density. At a density of 0.294 g·cm^{-3}, the compressive strength reaches up to 10.2 MPa. This result indicates that the PIHM has excellent compressive performance and is suitable for high–strength working environments.

3.5. Thermal properties of the PIHM

Fig. 3c shows the thermal conductivity of the PIHM with different ink viscosities and honeycomb side length. The average pore size of the PI aerogels remains at the micrometre level as the ice crystals grow smoothly during the freezing of the sol, and the growth is facilitated by the slippage between molecular chains. Moreover, high–viscosity ink restricts the growth of ice crystals, decreases the porosity of the PI aerogel and increases the solid–phase thermal conductivity, thus increasing the thermal conductivity coefficient. Thus, the thermal conductivity of the PIHM increases with the viscosity of the ink. Additionally, due to the reduced thermal conduction surface area and elongated conduction pathways of the thin walls within the honeycomb structure, the heat transfer in the solid phase is effectively suppressed. As the

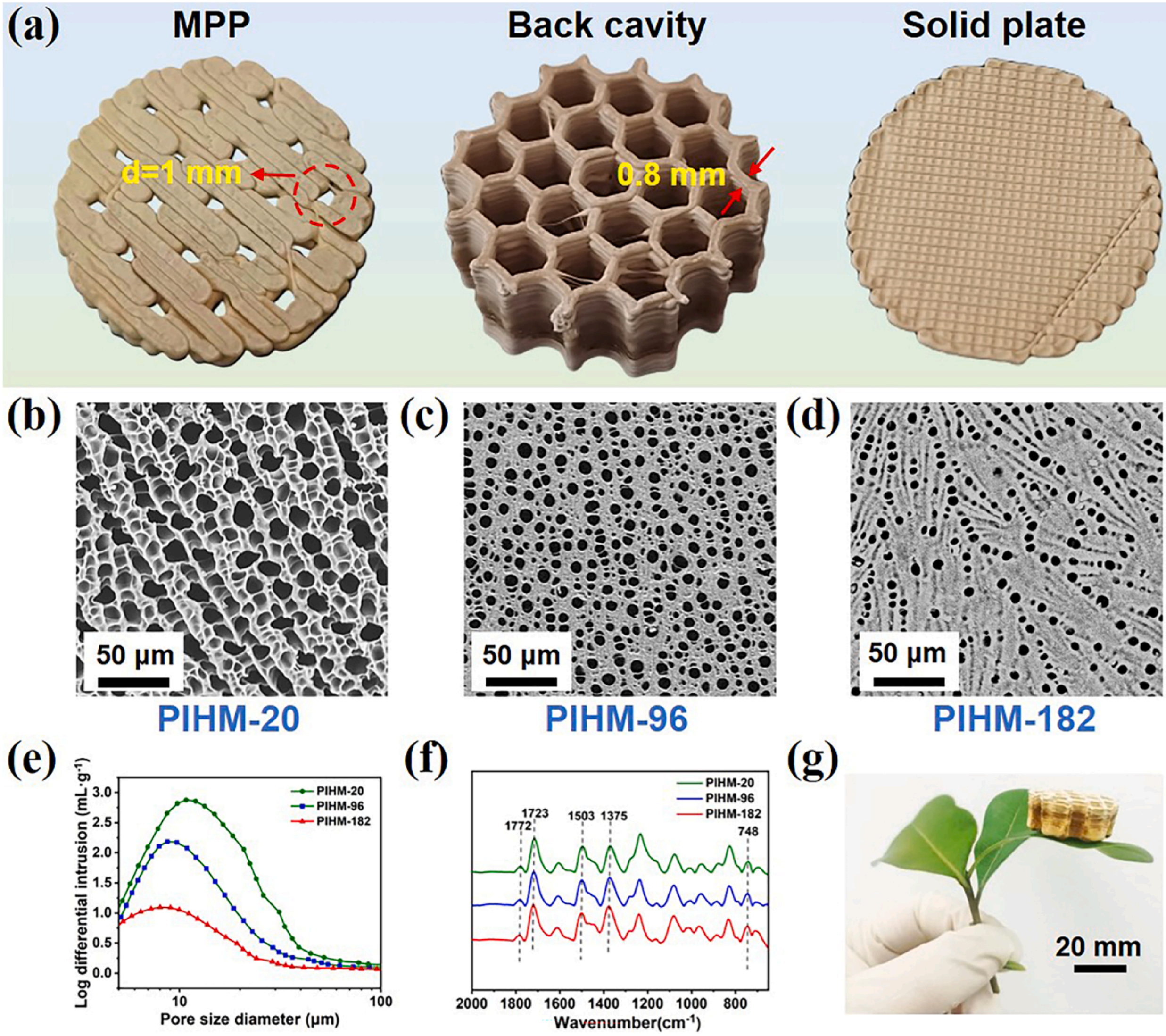

Fig. 2. Structural and density characteristics of PIHM. (a) Sample image of the compositional structure of PIHM. (b–d) SEM images of aerogels at different ink viscosities. (e) Pore size distribution curve and (f) FTIR spectra of PIHM at different ink viscosities. (g) Optical image of PIHM with a diameter of 30 mm and height of 15 mm, at an ink viscosity of 182 Pa·s, placed on green leaves.

Table 1
Typical physical properties of PIHM.

Sample	γ[a] (Pa·s)	L[b] (mm)	λ[c] ($W \cdot m^{-1} \cdot K^{-1}$)	ρ[d] ($g \cdot cm^{-3}$)	ξ_1[e] (%)	ξ_2[f] (%)
PIHM–1	20	2	0.0426±0.0011	0.206	16.2	22.6
PIHM–2	20	3	0.0373±0.0022	0.171	16.2	22.1
PIHM–3	20	4	0.0317±0.0008	0.153	16.2	21.5
PIHM–4	96	2	0.0566±0.0028	0.238	16.4	23.3
PIHM–5	96	3	0.0512±0.0009	0.201	16.4	22.8
PIHM–6	96	4	0.0455±0.0023	0.182	16.4	22.4
PIHM–7	182	2	0.0665±0.0012	0.294	16.6	24.6
PIHM–8	182	3	0.0618±0.0025	0.259	16.6	24.2
PIHM–9	182	4	0.0559±0.0015	0.225	16.6	23.7

[a] Ink viscosity.
[b] Honeycomb side length.
[c] Thermal conductivity.
[d] bulk density.
[e] Axial shrinking ratio.
[f] Radial shrinkage ratio.

honeycomb side length increases, the relative surface area of the solid material decreases, thereby reducing the avenues for heat conduction through the solid, leading to a reduction in solid–phase heat transfer. Conversely, the material's porosity increases, providing a larger space for gases to have more opportunities to transfer heat through natural convection, thereby enhancing gas–phase heat transfer. However, the convective heat transfer of gases is typically less efficient than the conductive heat transfer of solids. When the honeycomb side length is increased from 2 mm to 4 mm, the overall insulating capability tends to improve. When the ink viscosity is 20 Pa·s and the honeycomb side length is 4 mm, the thermal conductivity of the back cavity drops to 0.0317 $W \cdot m^{-1} \cdot K^{-1}$, demonstrating the excellent thermal insulation performance of the PIHM. The thermal stability of PIHM in air was characterised through thermogravimetric analysis (TGA). The PIHM begins to show significant weight loss at temperatures of 472.7–506.8 °C (<5 wt%) and gradually decomposes, and organic matter burns until the weight loss rate approaches 100 % at 630 °C (Fig. 3d). Fig. 3e–f presents the temperature variation curves and infrared image of PIHM–5 compared with cement block and lightweight wood, with all samples being 30 mm in height. After thermal equilibrium, the surface temperature of the PIHM–5 is 57.6°C, significantly lower than that of the

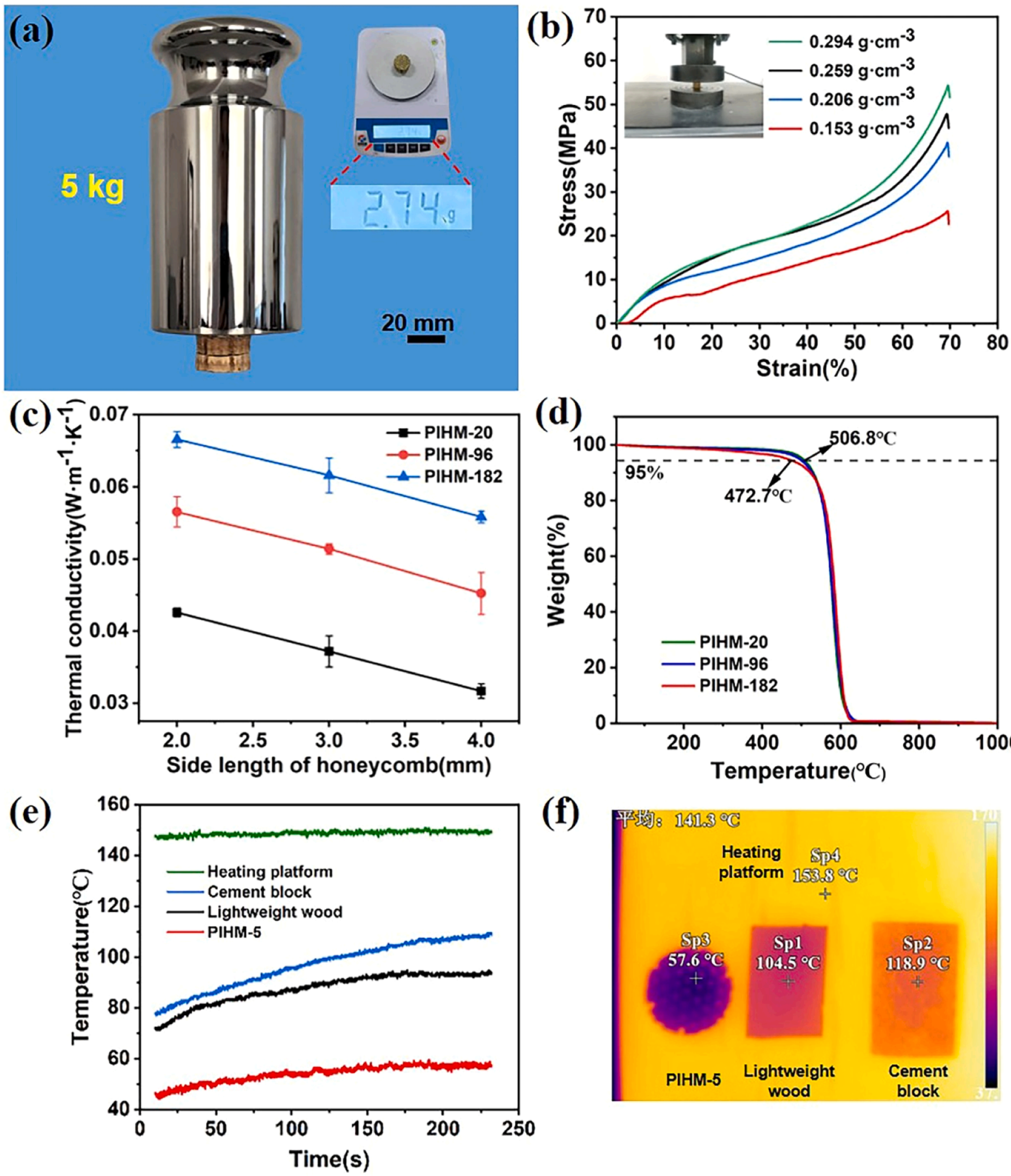

Fig. 3. Performance characterization of PIHM. (a) Optical image of a 2.74 g PIHM supporting a 5 kg weight. (b) Stress–strain curves of PIHM at different densities. (c) Variation curve of the thermal conductivity of PIHM with ink viscosity and honeycomb side length. (d) Thermal stability of PIHM with different ink viscosities in air. (e–f) The temperature variation curves and infrared image of PIHM–5, lightweight wood, and cement block rapidly placed on a 150°C heating platform.

cement block and lightweight wood, demonstrating superior insulating capability.

3.6. Sound–absorption performance of the PIHM

Sub–millimetre pores are characterised by low acoustic mass and high acoustic impedance. When sound waves enter through the MPP, frictional viscous loss occurs around the pores, converting a part of the sound energy into the dissipated heat energy. When the perforation diameter is sufficiently small, the dissipation effect is enhanced with increasing number of perforations or thickness of the perforated plate. Additionally, the combination of the MPP and back cavity can be regarded as an acoustic structure composed of multiple Helmholtz resonators in parallel. The sound–absorption mechanism is equivalent to that of a Helmholtz resonator. That is, the sound wave energy is consumed when the incident frequency of the sound wave matches the natural frequency of the structure. Moreover, when the sound waves that enter the cavity interact with the wall surfaces, secondary dissipation occurs within the aerogel pores, resulting in excellent sound–absorption performance (Fig. 4). Fig. 5a–d display the sound–absorption performance of the PIHM under different ink viscosities, back cavity depth, MPP thickness and honeycomb side length. Owing to the increased ink viscosity leading to a decrease in aerogel porosity, the sound waves entering the pore walls are reduced, the frictional viscous effect in the back cavity channels weakens and sound–absorption capability decreases. When the ink viscosity is

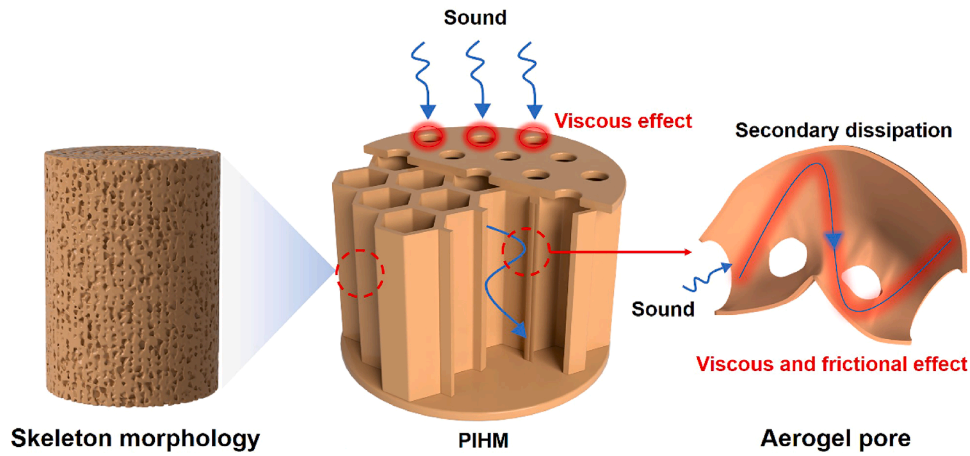

Fig. 4. The sound absorption mechanism diagram of PIHM.

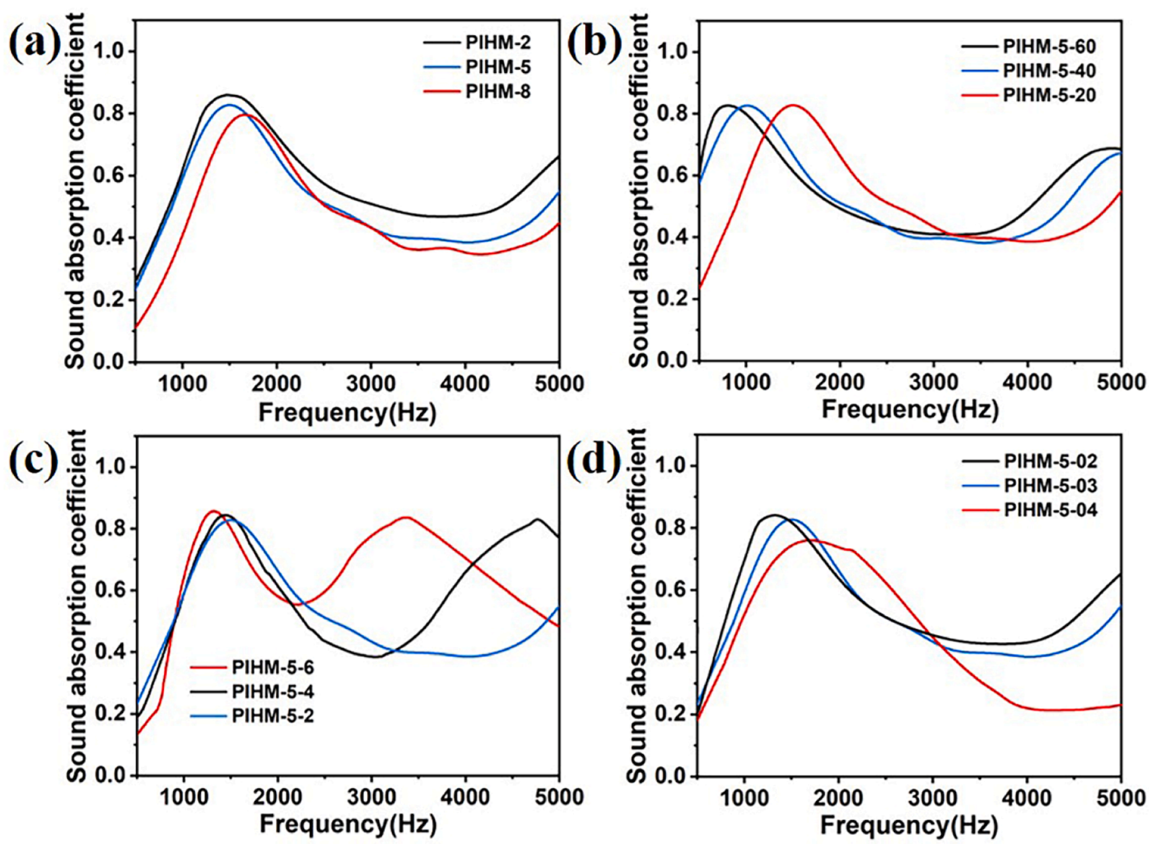

Fig. 5. Sound absorption performance of PIHM. (a) Sound absorption curves of PIHM with different ink viscosities, (b) different back cavity depths, (c) different MPP thicknesses, and (d) different honeycomb side lengths.

20 Pa·s, the resonance frequency is 1475 Hz, and the resonance absorption peak is 0.87. As the ink viscosity increases to 182 Pa·s, the resonance frequency increases to 1687 Hz, and the corresponding sound–absorption peak decreases to 0.79. The bandwidth of the sound–absorption coefficient greater than 0.8 slightly decreases with increasing viscosity (Fig. 5a). Therefore, the porous structure of the aerogel can convert sound energy into thermal energy, thus consuming it and affecting the resonance absorption peak and sound–absorption bandwidth. Additionally, frameworks with higher porosity exhibit lower resonance frequencies.

The most common method for reducing the resonance frequency of a structure is to increase the thickness of the sound–absorbing structure. According to the formula from MPP acoustic theory [19], the resonance frequency f of the MPP sound–absorbing structure can be considered the intersection of a cotangent function and a directly proportional function with respect to f. As the thickness of the back cavity increases, the cotangent function curve contracts overall towards the y axis along the x axis, whereas the directly proportional function with respect to f remains unchanged, and thus the intersection point shifts to the left, and the resonance frequency decreases. As shown in Fig. 5b, when the ink viscosity is 96 Pa·s and the back cavity thickness increases from 20 mm to 60 mm (PIHM–5–20, PIHM–5–40 and PIHM–5–60), the resonance frequency decreases from 1492 Hz to 790 Hz. However, the contraction of the cotangent function causes a leftward shift at the intersection, and the extent of the shift decreases with increasing back cavity thickness. Therefore, as the thickness of PIHM–5 increases from 20 mm to 60 mm, the extent of the leftward shift in the resonance frequency decreases. Coupled with the limitations of structural length and production costs, the method for obtaining low–frequency sound absorption by increasing the back cavity thickness has limitations.

Moreover, the acoustic theory formula for the MPP indicates that the relative acoustic impedance and relative mass reactance of the MPP sound–absorbing structure are directly proportional to the thickness of the perforated plate. When the back cavity thickness is 20 mm, and the honeycomb side length is 3 mm, increasing the perforated plate thickness from 2 mm to 6 mm (PIHM–5–2, PIHM–5–4 and PIHM–5–6) results in an increase in the resonance absorption peak of PIHM from 0.83 to 0.86. The corresponding resonance frequency decreases from 1492 Hz to 1323 Hz (Fig. 5c), and the sound–absorption bandwidth near the resonance peak gradually narrows. Correspondingly, when the back cavity thickness is 20 mm and perforated plate thickness is 2 mm, reducing the honeycomb side length from 4 mm to 2 mm (PIHM–5–04, PIHM–5–03 and PIHM–5–02) increases the resonance absorption peak of the PIHM from 0.76 to 0.84, and the resonance frequency shifts to left from 1673 Hz to 1309 Hz (Fig. 5d). The PIHM with a honeycomb side length of 4 mm exhibits a wide sound–absorption bandwidth near the peak because the small honeycomb side length results in additional micro–perforations. This feature leads to considerable scattering and to the absorption of the incident sound waves, thereby increasing the resonance absorption peak. Additionally, the PIHMs with small honeycomb side length have additional framework structures, increasing the area for frictional dissipative interaction between the sound waves and cavity walls. Conversely, a large honeycomb side length results in the low relative mass reactance and acoustic impedance of the MPP, leading to a broad sound–absorption bandwidth. Thus, the PIHM exhibits good mid–frequency sound–absorption performance. By increasing the back cavity and perforated plate thickness and reducing the honeycomb side length, the resonance absorption peak and resonance frequency can be lowered. However, the resulting improvement in performance is limited and inhibits the broadening of the absorption bandwidth. Achieving effective sound absorption across a wide and low–frequency range simultaneously remains challenging.

3.7. Sound–absorption performance of the PIHM acoustic metamaterial

By using the aerogel pore size and porosity as variables, a building block–assembly strategy is employed to serially connect two PIHMs (each with a cavity depth of 30 mm, honeycomb side length of 3 mm and perforated plate thickness of 2 mm). Given that changes in pore size or porosity changes in the two–layer assembly structure can be considered linear, the overall effect on sound waves is the same, making the arrangement order of the PIHMs irrelevant. Furthermore, as the aerogel pore size decreases from 11.2 μm (porosity of 82 %) to 5.2 μm (porosity of 60 %), the sound–absorption performance of the PIHM decreases accordingly. Therefore, to achieve optimal sound–absorption performance, the combination of acoustic metamaterials is as follows: PIHM–10 has an aerogel pore size of 11.2 μm; PIHM–11 is assembled from PIHMs with pore sizes of 11.2 and 8.6 μm (with a porosity of 70 %); PIHM–12 is constructed by joining PIHMs with pore sizes of 11.2 and 5.2 μm; and PIHM–13 is composed of PIHMs with pore sizes of 8.6–5.2 μm. Compared with the performance improvement attributed to increase in back cavity thickness, the serial combination of the PIHMs offers better

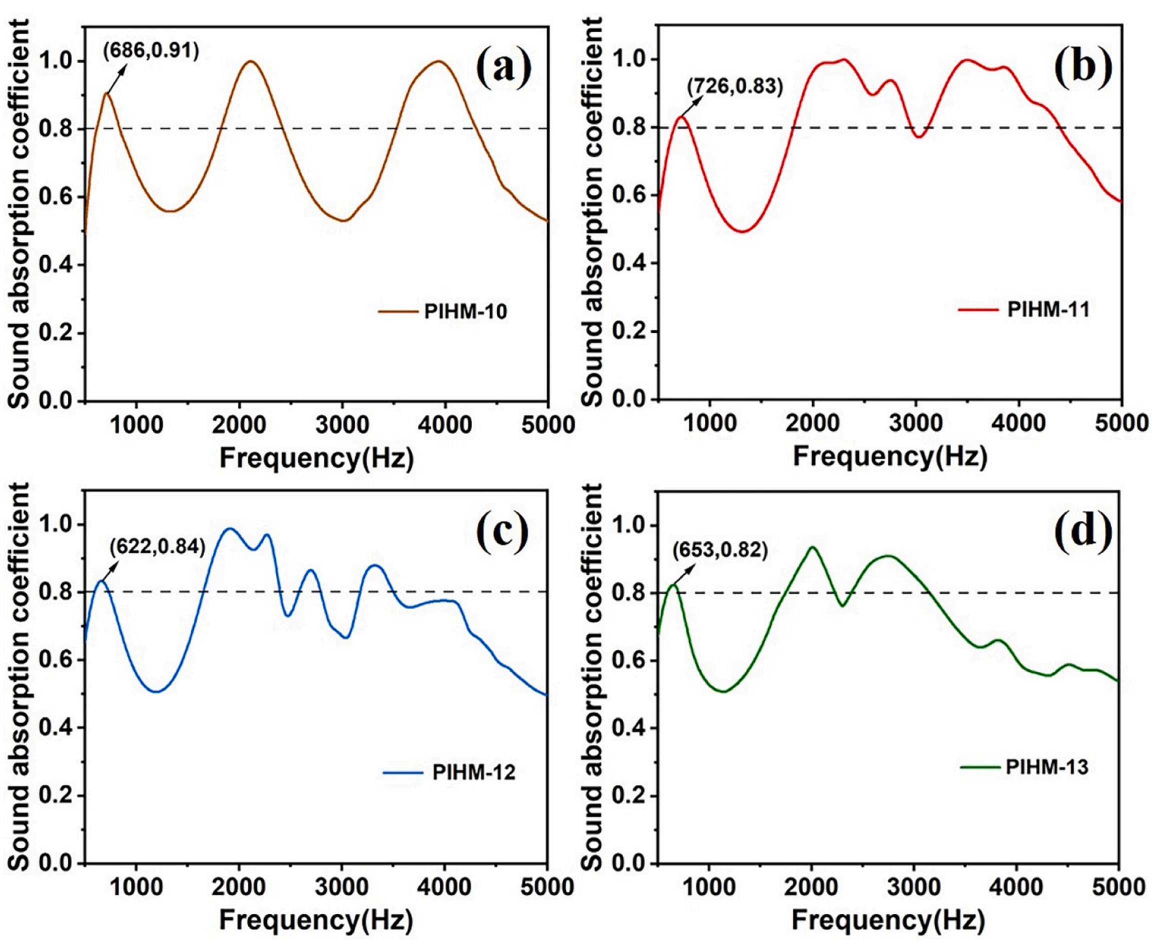

Fig. 6. Sound absorption performance of four PIHM acoustic metamaterials. (a) Sound absorption curve of PIHM–10, (b) PIHM–11, (c) PIHM–12, and (d) PIHM–13.

sound–absorption effects. The reasons are that the serial arrangement not only increases the length of the structure and reduces the resonance frequency and the second layer of the MPP has a certain relative mass resistance and acoustic impedance, causing loss to incoming sound waves. Consequently, this enhances the sound–absorption coefficient and shifts the resonance frequency further to the left. Therefore, the first resonance frequency of the serially connected PIHM structure is lower than that of PIHM–5–60, and the bandwidth of the sound–absorption coefficient greater than 0.8 is broadened. PIHM–10 has a sound–absorption peak of 0.91 at 686 Hz (Fig. 6a). PIHM–11 reaches a peak of 0.83 at 726 Hz (Fig. 6b). The first resonance frequency of PIHM–12 is 622 Hz, with a corresponding peak of 0.84 (Fig. 6c). PIHM–13 has a first resonance frequency of 653 Hz, with a sound–absorption peak of 0.82 (Fig. 6d). The sound–absorption performance of all serial structures considerably improves. Compared with the other three serial structures, PIHM–10 has a narrower sound–absorption bandwidth because PIHMs with the same aerogel pore size have similar natural frequencies, which have a minor effect on resonance frequency. Conversely, the serial sound–absorbing structures with varying aerogel pore sizes (PIHM–11, PIHM–12 and PIHM–13) effectively absorb sound waves across a broader range of frequencies, resulting in a wider sound–absorption bandwidth. However, a larger aerogel porosity leads to loss within the cavity during sound wave resonance, causing PIHM–10 to have a higher resonance peak than the other three sound–absorbing structures. Meanwhile, PIHM–13 exhibits the lowest sound–absorption coefficient among the structures with varying aerogel pore sizes. Thus, the acoustic metamaterial of PIHM not only can reduce the resonance frequency and increase the sound–absorption coefficient but also can effectively broaden the sound–absorption bandwidth. Table 2 compares the sound absorption frequency bands (absorption coefficients greater than 0.8) of PIHM-11 with various sound absorbing and thermal insulation materials, demonstrating that the PI aerogel metamaterials possess superior noise reduction characteristics.

3.8. *Application of the PIHM and acoustic metamaterial in the field of architecture*

Fig. 7 compares the fundamental physical properties of PIHM with those of common architectural sound–absorbing and thermal insulation materials. The comparison reveals that PIHM is lightweight, provides thermal insulation and possesses exceptional compressive strength. Notably, compared with 3003 aluminium honeycomb panels possessing a similar honeycomb structure, a PIHM not only has a lower density and higher compressive strength but also offers superior thermal insulation capabilities. Moreover, the aerogel framework structure endows a PIHM with enhanced sound–absorption performance. Relative to high–porosity wood, a PIHM's primary advantage lies in its remarkable mechanical strength, which contributes to robust and stable construction. Furthermore, serially connecting PIHMs with varying aerogel pore sizes facilitates broadband and low–frequency sound absorption. The

Table 2
Comparison of acoustic properties between PI aerogel metamaterials and thermal insulation and sound absorption materials.

Material	Size (mm)	Frequency (Hz)	Ref.
PIHM–11	60	650–800 1813–2950 3124–4934	
Expanded vermiculite	50	1100–1400	[20]
Expanded pearlite	50	700–1400	[20]
Fiber glass	50	500–1200	[20]
Ultra-lightweight concrete	80	800–900	[21]
Polyurethane foam	95	1200–4700	[22]
PI/aramid sponge	50	1900–4400	[23]

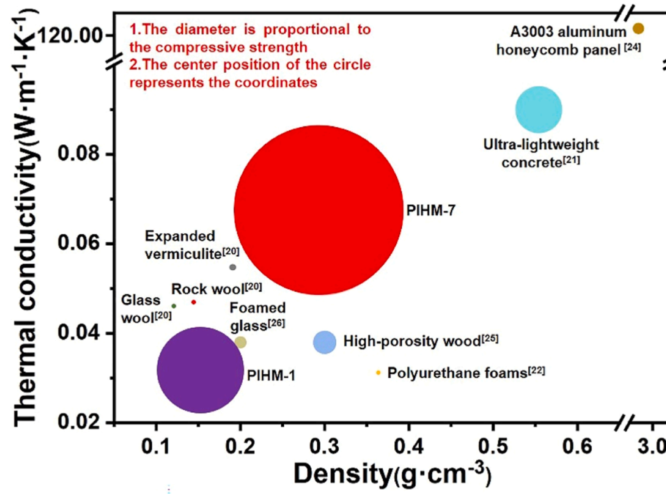

Fig. 7. Performance comparison of PIHM with commonly used architectural sound absorbing and thermal insulation materials in terms of density, thermal conductivity, and compressive strength. ([24–26]).

application spectrum of the sound–absorbing structure can be broadened by substituting materials for the MPP and base plate. As illustrated in Fig. 8, a 2 mm–thick aluminium plate is fashioned into a MPP and solid base plate, which are then bonded to a PIHM back cavity of 30 mm thickness and 70 mm diameter with a high temperature–resistant adhesive. The resulting durable and compression–resistant aluminium/PIHM composite sound–absorbing panel (60.28 g) is suitable for outdoor environments, such as houses, malls and skyscrapers. Similarly, the lightweight wood/PIHM composite sound–absorbing panel (67.47 g) offers enhanced thermal insulation properties, which makes it ideal for indoor venues, such as cinemas, concert halls and conference centres. Moreover, both types of composite panels can withstand external pressure 1555 times their own weight, effectively resisting external impacts and indicating the structure's robustness against failure. Hence, the PIHM exhibits strong versatility and practicality.

4. Conclusion

The PIHM fabricated through freeze casting–assisted DIW not only exhibits outstanding sound–absorption capabilities but also retains the lightweight and thermal–insulating properties of aerogels. Furthermore, its high ink viscosity and hexagonal honeycomb structure contribute to a compressive strength of up to 10.2 MPa (with a density of 0.294 g·cm^{-3}). Compared with conventional architectural sound–absorbing and thermal insulation materials, the PIHM demonstrates exceptional performance and can be assembled with aluminium and wood to form composite sound–absorbing panels, showcasing strong practicality and versatility for potential applications in construction, military and transportation sectors. Additionally, an aerogel–based acoustic metamaterial is developed by controlling ink viscosity and employing a building block–assembly strategy for serial connection. The fabricated aerogel–based acoustic metamaterial demonstrates broadband and low–frequency sound absorption. This work serves as an important reference for the research of 3D–printed aerogel metamaterials in construction.

CRediT authorship contribution statement

Jun Chen: Data curation. **Kunfeng Li:** Methodology. **Zichun Yang:** Supervision, Conceptualization. **Yan Gui:** Writing – original draft, Visualization, Methodology, Investigation. **Zhifang Fei:** Writing – review & editing, Investigation, Funding acquisition. **Shuang Zhao:** Visualization, Supervision, Funding acquisition. **Zhen Zhang:**

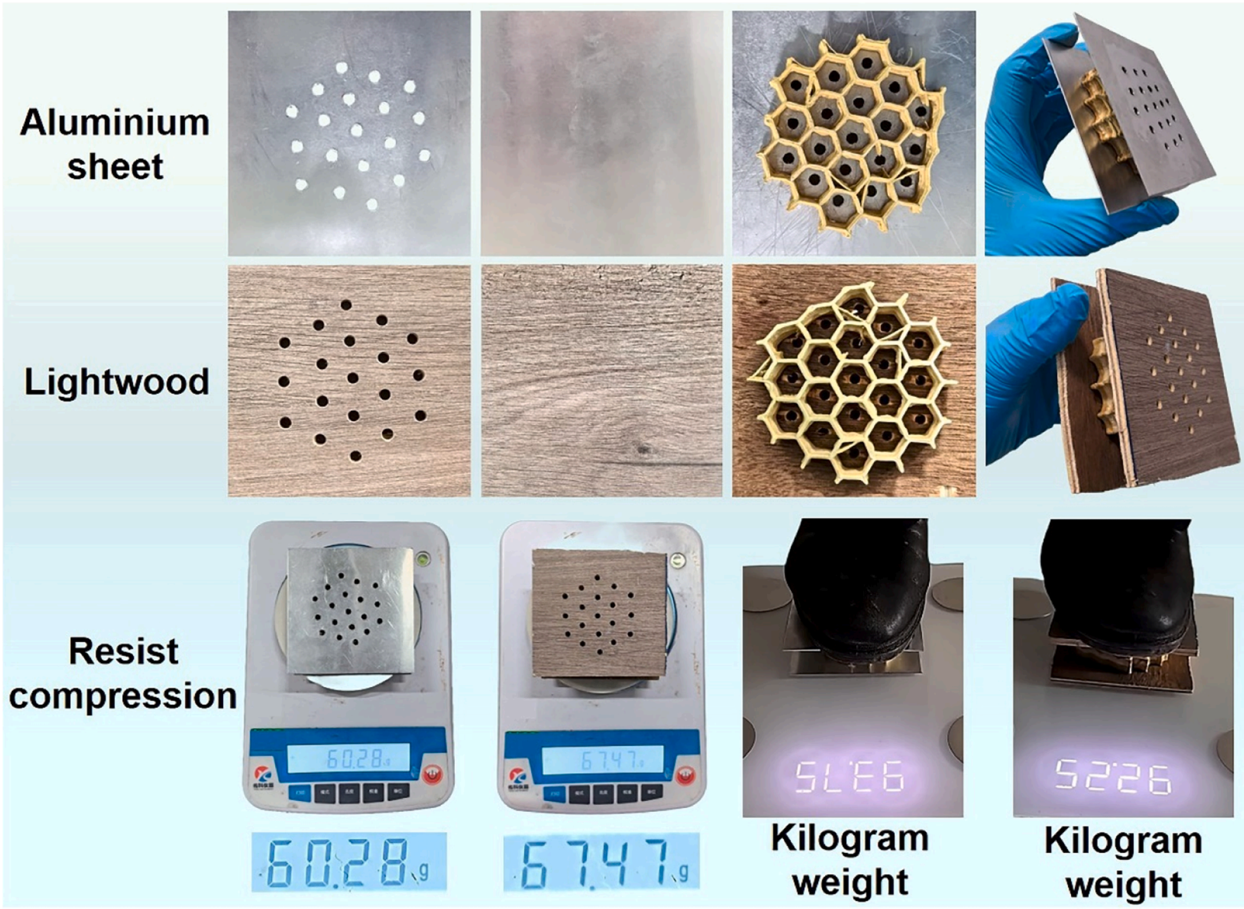

Fig. 8. Two kinds of self–made PIHM composite sound absorbing boards with high strength.

Methodology, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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